

LUSAKA, Zambia

WMO: 676650

Lat: 15.333S

Lon: 28.450E

Elev: 1152

StdP: 88.23

Time Zone: 2.00 (E02)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	8.1	-1.2	3.9	21.8	0.9	4.6	20.2	9.1	19.3	8.3	19.6	1.5	120	0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	14.8	34.1	18.3	33.0	18.6	31.9	18.5	24.4	26.4	23.7	26.0	23.2	25.6	3.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	21.8	25.4	23.2	20.7	24.8	22.7	20.0	24.4	81.7	26.6	78.3	26.6	75.7	25.4	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	7.2	6.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.0	22.8	22.8	22.2	20.8	18.3	16.5	16.1	18.4	22.0	24.9	24.5	23.3
	DBStd	3.38	1.08	1.14	1.39	1.79	1.53	1.84	1.79	2.10	2.19	2.10	2.08	1.51
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	181	0	0	0	2	20	61	72	24	2	0	0	0
	CDD10.0	4033	398	359	379	325	258	194	189	262	361	461	434	413
	CDD18.3	1172	140	126	121	77	19	5	3	28	113	203	184	154
	CDH23.3	9383	603	556	619	481	267	143	138	504	1299	2331	1569	872
(14) Wind	WSAvg	2.6	1.9	2.2	2.3	2.8	2.8	2.6	2.9	2.9	3.0	3.0	2.6	2.1
	PrecAvg	792	216	154	89	25	3	0	0	0	1	13	85	191
	PrecMax	1052	431	334	207	152	31	2	0	1	9	56	212	356
	PrecMin	446	86	57	0	0	0	0	0	0	0	0	35	68
	PrecStd	175	82	72	59	38	8	0	0	0	2	15	35	83
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.8	30.9	31.2	29.9	28.7	27.9	28.1	30.9	34.0	36.6	34.8	32.2
		MCWB	19.9	20.4	19.2	19.6	17.6	16.5	16.6	17.0	16.5	17.7	19.0	19.0
	2%	DB	29.1	29.0	29.1	28.6	27.2	26.2	26.3	29.1	32.8	35.0	33.3	30.7
		MCWB	21.6	21.7	21.5	20.7	18.1	16.1	15.2	16.0	17.5	18.1	19.1	19.9
	5%	DB	28.1	27.8	28.0	27.7	26.2	25.1	25.0	28.0	31.2	33.8	32.1	29.1
		MCWB	21.7	22.2	21.8	20.6	18.1	15.9	14.8	15.8	17.2	18.4	19.3	21.0
	10%	DB	26.9	26.8	26.9	26.5	25.0	23.9	23.7	26.5	30.1	32.5	30.8	27.8
		MCWB	21.5	21.9	21.3	20.1	17.6	15.2	14.2	15.5	16.8	18.5	19.3	21.1
	0.4%	WB	25.1	25.9	25.2	23.9	22.2	19.7	19.4	20.6	21.8	22.2	23.9	24.3
		MCDB	27.3	27.4	27.3	26.8	25.7	24.6	26.2	27.3	31.2	30.7	26.6	26.8
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	23.9	24.7	23.7	22.8	20.5	18.6	17.0	18.1	19.8	21.0	23.0	23.7
		MCDB	26.2	25.9	26.3	25.9	24.6	23.6	24.0	26.2	29.1	30.7	25.9	26.3
	5%	WB	23.3	23.8	22.9	21.7	19.3	17.5	15.8	17.2	18.6	20.2	22.3	23.0
		MCDB	25.6	25.2	25.9	25.5	24.1	22.6	23.0	25.4	28.6	30.1	25.8	25.7
	10%	WB	22.7	23.0	22.1	20.9	18.3	16.2	15.0	16.2	17.5	19.6	21.5	22.1
		MCDB	25.0	24.6	24.9	24.7	23.3	21.8	21.9	24.6	27.9	29.5	25.8	24.7
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.4	9.6	10.8	13.7	14.6	14.9	15.6	15.5	14.8	11.2	8.9
		MCDBR	10.7	10.3	11.4	12.4	14.9	15.9	17.1	17.5	17.0	16.5	14.0	11.4
		MCWBR	4.9	5.1	5.9	5.8	7.6	7.5	8.2	7.2	7.1	6.5	5.6	4.8
	5% WB	MCDBR	8.4	8.3	9.5	10.9	13.7	14.2	15.3	15.6	15.8	14.8	10.2	8.7
		MCWBR	5.4	6.0	6.4	6.5	8.6	8.3	8.8	8.3	7.9	6.9	5.3	5.0
(40) Clear-Sky Solar Irradiance	taub		0.376	0.376	0.357	0.314	0.295	0.294	0.317	0.372	0.452	0.438	0.404	0.381
	taud		2.458	2.471	2.524	2.621	2.619	2.566	2.473	2.284	2.085	2.159	2.309	2.434
	Ebn at Noon		968	959	957	966	953	936	919	892	853	891	934	961
	Edh at Noon		120	118	108	93	87	88	99	128	166	159	139	123
(44) All-Sky Solar Radiation	RadAvg		5.38	5.48	5.45	5.44	5.32	4.85	5.02	5.83	6.46	6.75	6.17	5.54
	RadStd		0.51	0.58	0.46	0.42	0.24	0.24	0.29	0.17	0.20	0.35	0.35	0.45

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page